Reg. No. | | | | | | | | | | |

[14ME46]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE – 2016

APPLIED THERMODYNAMICS

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions. Choosing at least two questions form each part.
2. Missing data may be suitably assumed.
3. Use of thermodynamics tables and charts are permitted

PART-A

1. a. An engine working on Otto cycle in which the salient points are 1,2,3,4, as upper and lower temperature limits are T_3 and T_1. If the maximum work per kg of air is to be done, then show that the intermediate temperature is given by $T_2=T_4=\sqrt{T_1T_3}$ **06**

b. The Diesel cycle has compression ratio 16. Temperature before compression is 300K. temperature after expansion 900K, determine: **08**

i. Net work output per kg mass of air

ii. Thermal efficiency

iii. Specific air consumption in terms of kg/kW-hr

2. a. Define: **05**

i. Stoichiometric air

ii. Enthalpy of formation

iii. Enthalpy of reaction

iv. Adiabatic flame temperature

v. Excess air

b. The products of combustion of an unknown hydrocarbon is C_x H_y have the following composition as measured by an Orsat apparatus CO_2=8.0% CO=0.9% O_2=8.8% and N_2=82.3% determine **09**

i. Composition of the fuel

ii. The air fuel ratio

iii. The percentage of excess air

3. a. Explain the different methods of importing the efficiency of Brayton cycle. **06**

b. In an open cycle constant pressure gas turbine air enters the compressor at 1 bar and 300K. Pressure of air after compression is 4 bar. Isentropic efficiency of compressor and turbine are 78% and 85% respectively. Air fuel ratio is 85:1. Find input power and thermal efficiency of the cycle. If the flow rate of air is 25kg/s Cp=1kJ/kgK and γ=1.4 for air, Cpg=1.147 kJ/kg K and γ_g=1.33 for gases. R =0.287kJ/Kg K, calorific value of fuel is 42000 kJ/kg **08**

4. a. Sketch the flow diagram and corresponding temperature entropy diagram of a reheat vapour cycle and derive an expression for the reheat cycle efficiency. **06**

b. A 40 MW steam power plant working on Rankine cycle operates between boiler pressure 4MPa and condenser pressure of 10kPa. The steam leaves the boiler and enters the steam turbine at 400^{0}C. The isentropic efficiency of the steam turbine is 85%. Determine: **08**

i. the cycle efficiency

ii. the quality of steam from the turbine

iii. the steam flow rate in kg/hr considering pump work

PART-B

5. a. With a neat sketch, explain the working of vapor absorption refrigeration system. **06**

b. A Bell Coleman cycle works between 1 bar and 5 bar. The adiabatic efficiency of compression is 85% end expansion is 90%. Find COP of the system and its tonnage capacity when air flow rate is 1 kg/s. The ambient temperature is 27^{0}C and refrigerator temperature is 0^{0}C. **08**

6. a. Explain briefly the following frictional power determination methods **06**

i. Willian's line method

ii. Morse test method

b. A test on a single cylinder 4 stroke oil engine having bore of 15cm and stroke 30cm gave the following results: Speed=300 rpm, indicated mean effective pressure= 7 bar, brake torque=200N-m, cooling H_2O rise in temperature =30^{0}C, air fuel ratio=22:1. **08**

Exhaust gas temperature=410^{0}C, specific heat of exhaust gas = 1kJ/kgK CV of fuel =42 MJ/kg. R=0.2872 kJ/kg K, determine:

i. Indicated thermal efficiency

ii. volumetric efficiency based on atmospheric condition

iii. Draw an energy balance sheet in terms of kJ/min and on percentage basis.

7. a. Derive an expression for minimum work required by two stage air compressor **06**

with perfect intercooling between stages.

b. A single acting air compressor has a cylinder of bore 15cm and piston stroke is 25cm. the crank speed is 600rpm. Air is taken from atmosphere at 1 bar and 27^0C and delivered at 11 bar. Assume polytrophic compression of the type $PV^{1.25}=C$. Find power required to drive the compressor, if its mechanical efficiency is 80%. The compressor has a clearance which is $\frac{1}{20}^{th}$ of stroke volume. How long will it take to deriver $1m^3$ of air at compressor inlet conditions. Also find the volumetric efficiency of the compressor. **08**

8. a. With a neat sketch explain the working of air condition system for hot and dry weather. Present the processes involved on psychrometric chart. **06**

b. Moist air at a temperature of 21^0C under a total pressure of 736 mm of Hg. The DPT is 15^0C find. **04**

i. Partial pressure of H_2O vapour

ii. Relative humidity

iii. Specific humidity

iv. Enthalpy of air per kg of water vapour

v. Specific volume of air per kg of water vapour